Pidaparthi Durga Pranavi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.97
Sri Chaitanya College	Class XII (TSBIE)	2024	98.90%
Sri Chaitanya School	Class X (TS SSC)	2022	10.00%

PROJECTS

Multithreaded Document Search Engine

C++ , Information Retrieval

- Built a multithreaded document search engine implementing an inverted index with text tokenization and normalization, supporting BM25 and TF-IDF ranking for efficient information retrieval.
- Indexed 19,997 documents in ~8.4 seconds with Top-K retrieval, and microsecond-level query latency measurement.
- Implemented Boolean query parsing , exact phrase search, and BK-tree-based fuzzy search using Levenshtein edit distance for automatic spelling correction and improved search accuracy.

Multi-Store Price Monitoring & Alert System

Python , Playwright , BeautifulSoup , SQLite

- Developed a modular multi-store scraping system using Playwright and BeautifulSoup, supporting 4 e-commerce platforms scraping products with an average latency of ~5.3 s per product.
- Implemented a SQLite-based data management system to maintain current product states and historical price/stock trends, enabling change detection and price trend visualization using Matplotlib.
- Built a product alerting system with event-based email notifications and structured logging to detect price/stock changes .

xv6 Kernel Enhancements

C,xv6,Operating Systems

- Extended xv6 with custom system calls for PID/PPID retrieval, syscall counters, and process runtime statistics.
- Implemented a Multi-Level Feedback Queue scheduler with priority boosting and runtime metrics for improved process scheduling analysis.
- Enhanced virtual memory and storage management with demand paging, lazy allocation, page eviction, disk-backed swapping, RAID (0/1/5), and disk scheduling algorithms

Numerical Boundary Value Problem Solver

Python, NumPy, Matplotlib

- Developed a modular Boundary Value Problem solver implementing the Shooting Method and Finite Difference Method, with custom implementations of RK2, RK4, ABM2, ABM4, Newton's method, and the Thomas algorithm.
- Designed reusable numerical solver components for ODE integration, nonlinear system solving, and tridiagonal linear system optimization.
- Evaluated solver accuracy, stability, and convergence across multiple discretizations by benchmarking Shooting Method and FDM solutions .

TECHNICAL SKILLS

Languages : C++, C, Python

Libraries : NumPy, Pandas, Matplotlib , Playwright , BeautifulSoup

Tools : VS Code , Github, Jupyter Notebook , SQLite ,MySQL

RELEVANT COURSES

Computer Science : Introduction to Programming , Data Structures and Algorithms , Operating Systems I & II

Mathematics : Calculus , Differential Equations , Transform Techniques , Complex Variables , Probability , Introduction to Statistics

Scientific Computing : Computer Aided Numerical Methods I & II , Fundamentals of Scientific Computing , Computational Methods in Material Science

Electrical : Digital Circuits , Introduction to HDL

POSITIONS OF RESPONSIBILITY

Core Member , Blockspace - Cryptocurrency Club

- Collaborated with core members on blockchain discussions and club initiatives to promote learning within the student community.

Class Representative (2024-2025)

- Served as the primary liaison between faculty and students, streamlining communication and addressing academic issues.

Core Member , Shuffle -Dance Club

- Coordinated dance practices, performances, and club activities, contributing to the successful execution of institute cultural events.

EXTRACURRICULAR

- Represented IITH in the dance contingent at the Inter-IIT Cultural Meet.
- NSS Volunteer